Computer Network Practice: *GARP, PARP*

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Lecture Plan

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7	**10.13**	**GARP, RARP**
8	10.20	(middle test)
9	10.27	Internet Protocol Routing I & Static Router
10	11.3	Internet Protocol Routing II & Static Router
11	11.10	Internet Protocol Routing III & Static Router
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Gratuitous ARP





Gratuitous ARP

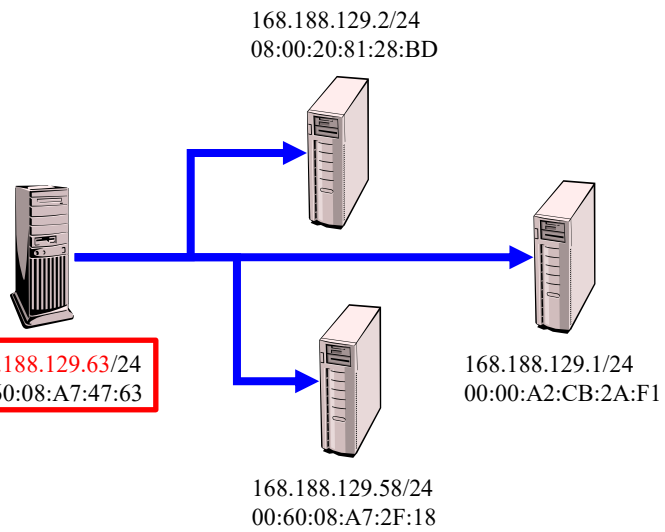
- It occurs when a host sends an ARP request looking for its own IP address.
- This is usually done when the interface is configured at bootstrap time.
- It lets a host determine if another host is already configured with the same IP address.
- If a reply is received, the error message “duplicate IP address sent from Ethernet address: xx:xx:xx:xx:xx:xx” is logged on the console.

Gratuitous ARP

- If the host sending the gratuitous ARP has just changed its hardware address.
  - the host was shut down, the interface card replaced, and then the host was rebooted
- Update old hardware address entry in ARP cache
  - if a host receives an ARP request from an IP address that is already in the receiver's cache,
  - then that cache entry is updated with the sender's hardware address from the ARP request.
- This is done for any ARP request received by the host.

Example of Gratuitous ARP

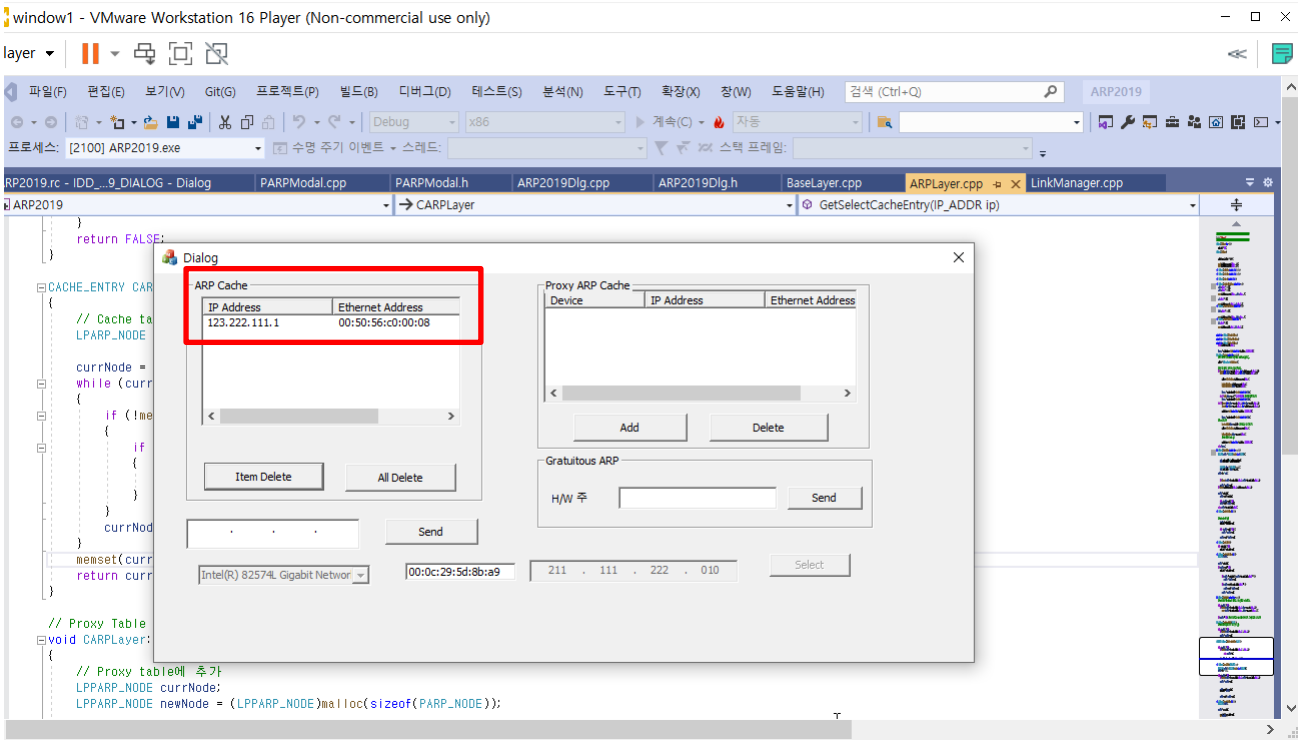
	ff:ff:ff:ff:ff:ff
	00:60:08:A7:47:63
	0x0806
Hardware type	1
Protocol type	0x0800
Length of hardware address	6
Length of protocol address	4
Opcode (ARP Request)	1
Sender's hardware address	00:60:08:A7:47:63
Sender's protocol address	168.188.129.63
Target's hardware address	???????
Target's protocol address	168.188.129.63



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Gratuitous ARP Operation

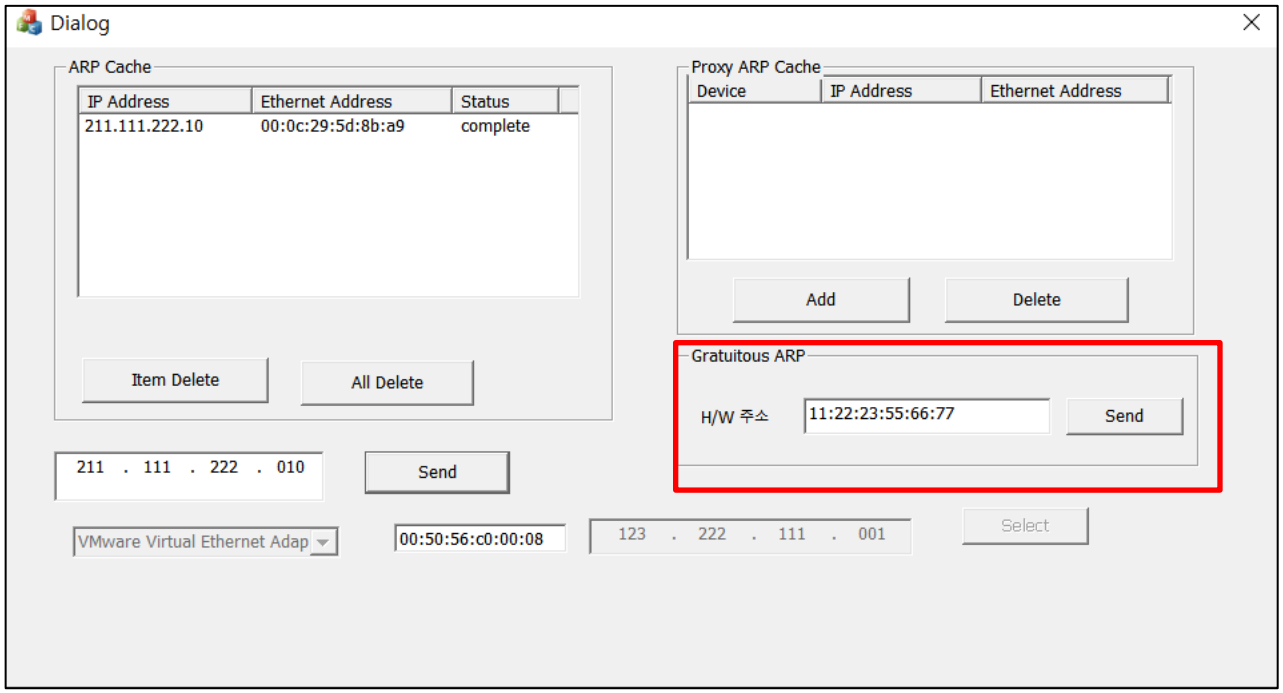
- Host B – There is information about Host A in Host B'S ARP Cache Table



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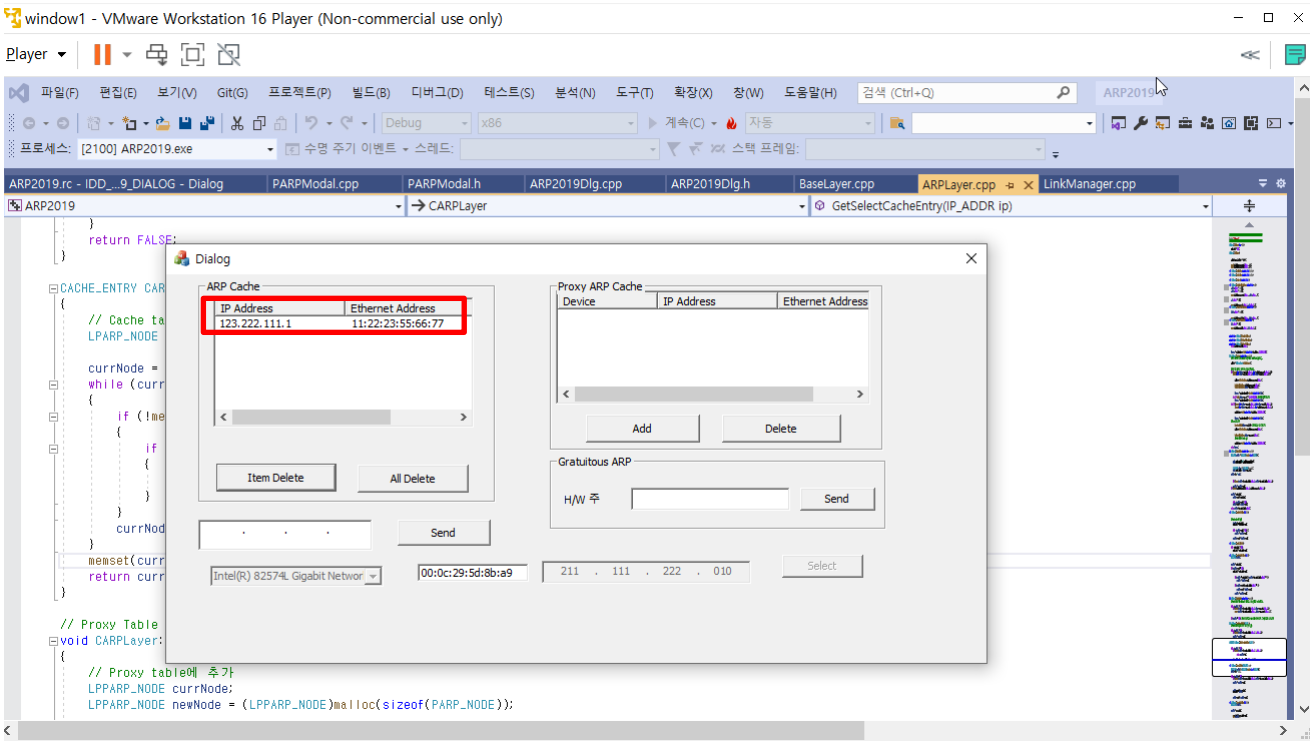
Gratuitous ARP Operation

- Host A – Send Gratuitous ARP Message to Host B



Gratuitous ARP Operation

- Host B – Update ARP Cache Table



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Proxy ARP

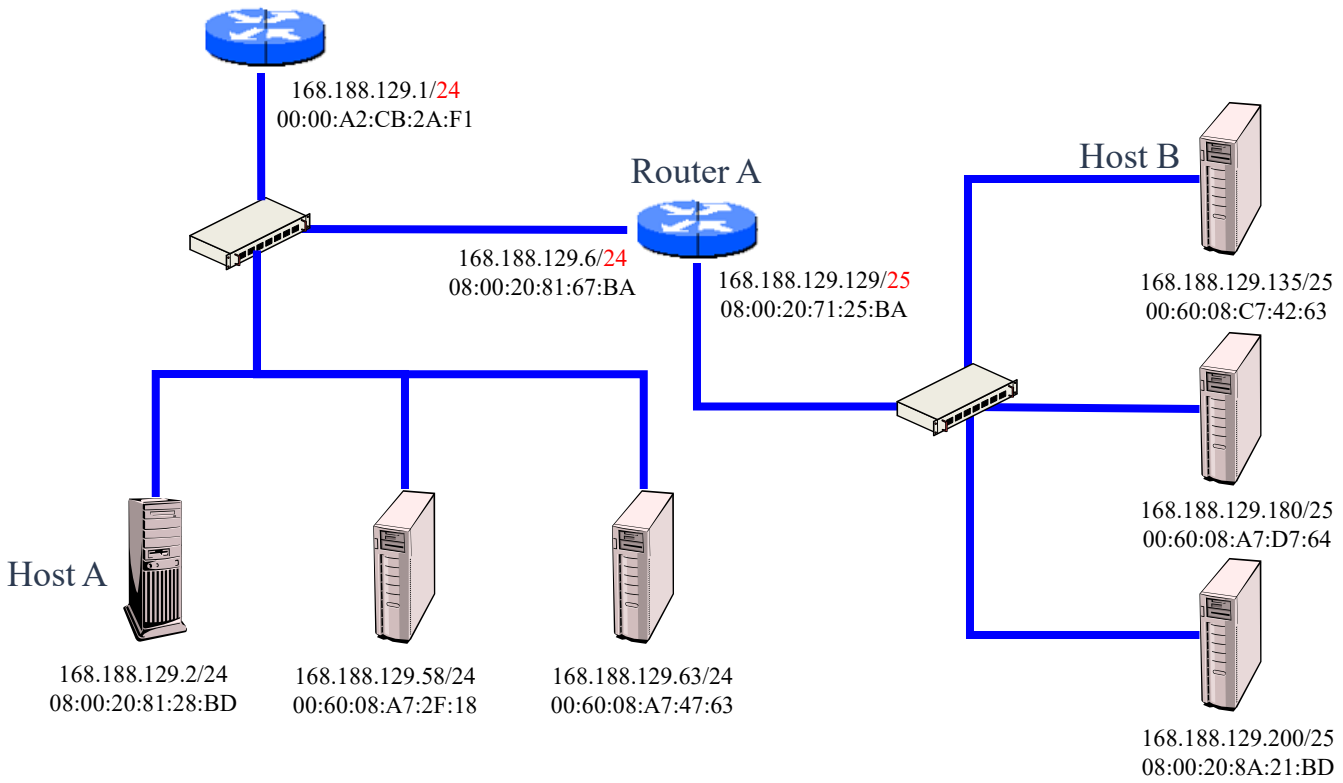




Proxy ARP

- Proxy ARP lets a router answer ARP requests on one of its networks for a host on another of its networks.
- Although in fact the destination host is on the other side of the router, this fools the sender of the ARP request into thinking that the router is the destination host.
- This router is acting as a proxy agent for the destination host, relaying packets to it from other hosts.

Example of Proxy ARP



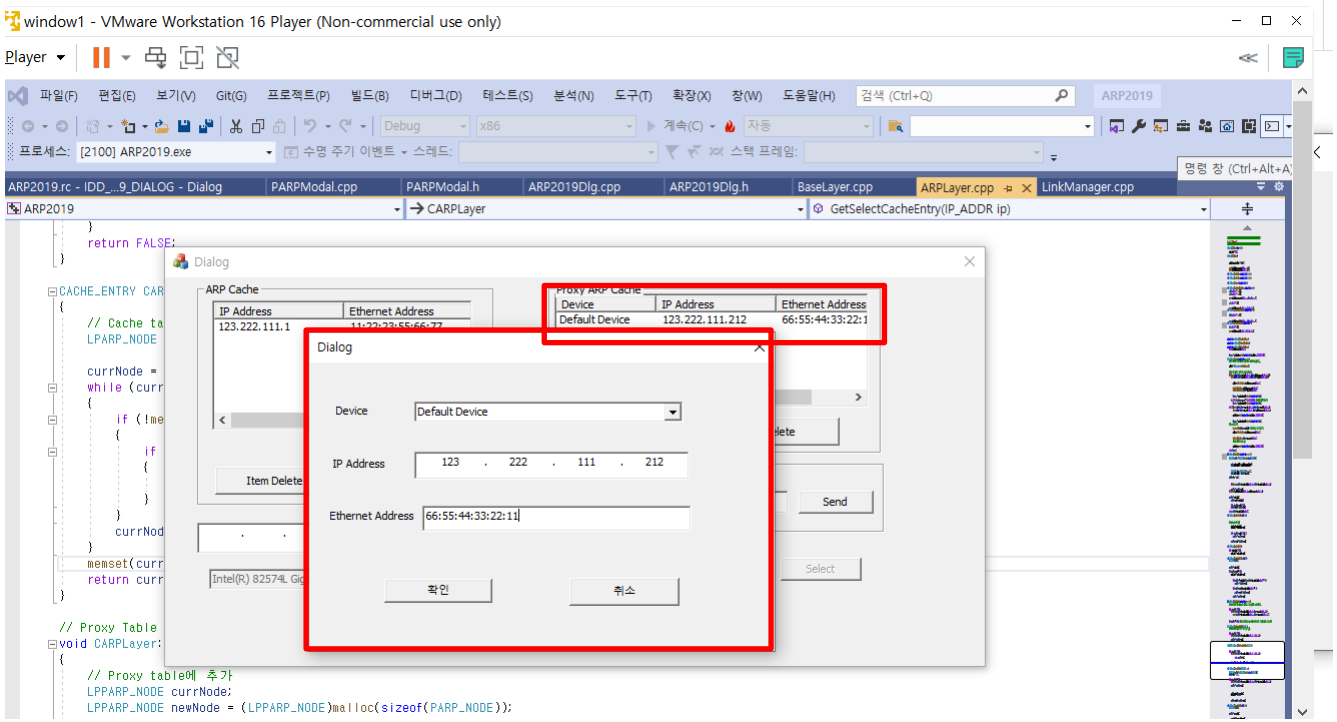
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Proxy ARP Operation

- Proxy ARP Operation
  - Executed ARP Requested Operation By Other Host
  - Proxy ARP Table
    - Entry Field
      - Device Name
      - IP Address
      - Ethernet Address
  - Wait for ARP Request Packet to arrive at My Host.
    - My Host address is not Target Protocol address in ARP Request Packet.
    - Search Proxy ARP Table.
      - If exists result entry? Then, Use those Entry to ARP Reply.
    - Append/Update a Entry to Cache.
    - Send to Reply Packet.

Proxy ARP Operation

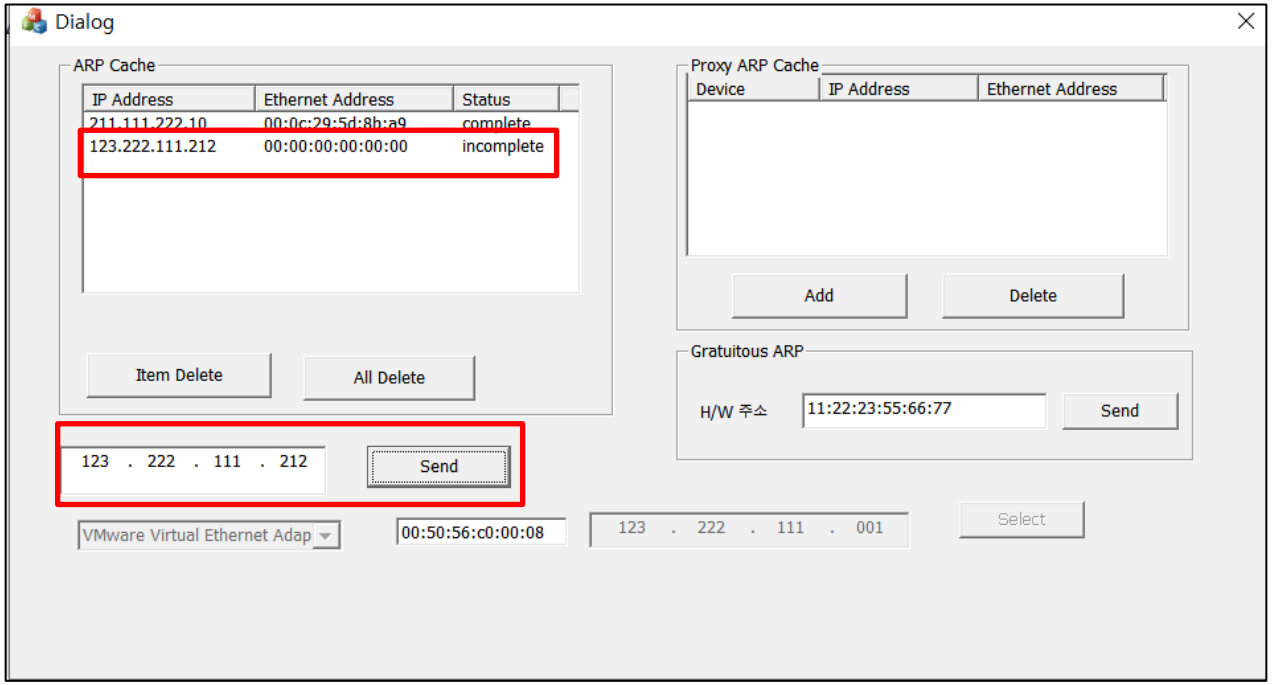
- Host B – Add Proxy ARP Entry



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Proxy ARP Operation

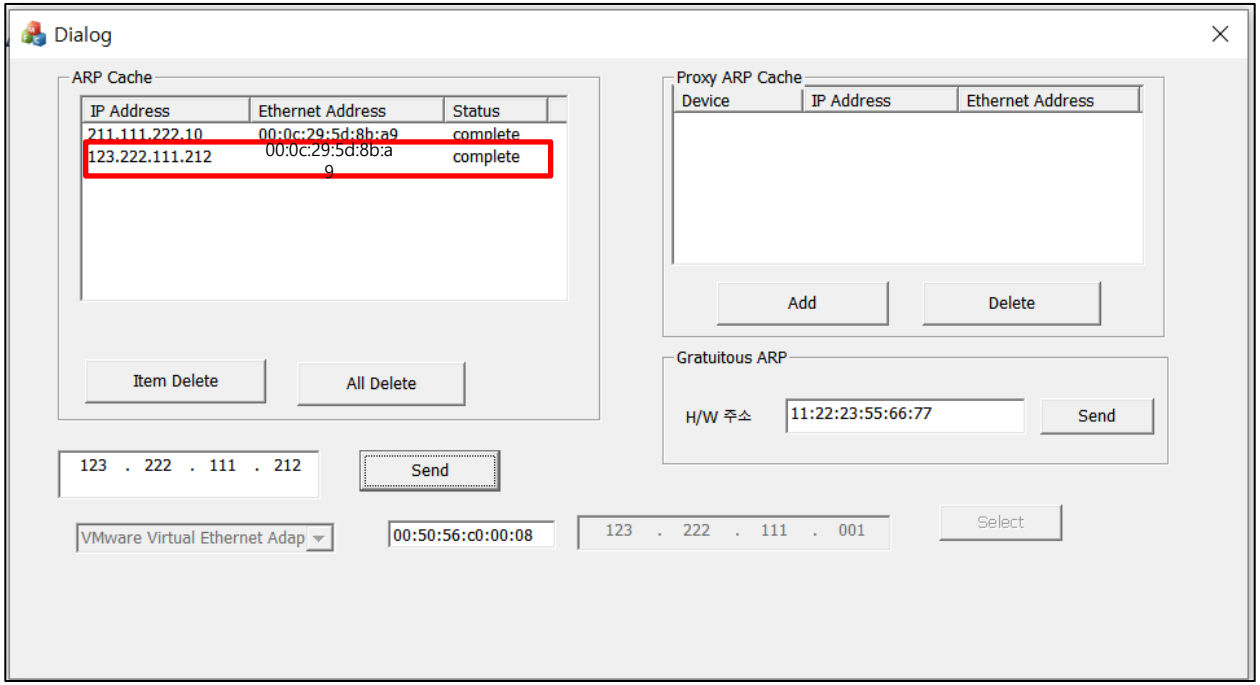
- Host A – Send ARP Request Message about Host B's IP Address



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Proxy ARP Operation

- Host A – Receive ARP Reply Message and Update ARP Cache Table



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Demo Notice

- 10월 17~18일(토, 일) 13:00 ~ 17:00 사이
  - 팀당 20분
  - 조 별 시간 지정은 구글 폼 선착순 접수
- 평가 주제
  - 팀별로 구현한 ARP 프로그램 기능(Basic ARP, GARP, PARP) 시연
  - 구현한 프로그램에 대해 설명할 수 있어야 함
- 장소
  - E8-1 424호
- 방법
  - 노트북 2개를 LAN선으로 연결하여 통신 확인
    - 팀으로 2대 모두 지참 가능
  - 노트북 1대는 제공할 수 있음 환경 설정 완료 상태로 제공
- 주의
  - 반드시 데모 시간 숙지
    - 시간 안에 데모 실패 시, 당일 마지막 순번으로 재배정
    - 반드시 스케줄 지킬 것

Q&A



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